



**Babu Madhav Institute of Information Technology**  
**Uka Tarsadia University**  
**Academic Year: 2026–2027**

# Smart Traffic Prediction System

*Subject: Cloud Computing*

## **Submitted By:**

Manasvi Limbasiya – 202504102510009(Msc IT)

Vaibhav Bagul -202206100110208

## **Guided By:**

*Dr.Bhumika Patel*

# INDEX

| No | Title                      | Page No |
|----|----------------------------|---------|
| 1  | Introduction               | 03      |
| 2  | Cloud Services Used        | 04      |
| 3  | Cloud Services Integration | 05      |
| 3  | Architecture Design        | 07      |
| 4  | Component Explanation      | 08      |
| 5  | System Work Flow           | 09      |
| 6  | Conclusion                 | 11      |

# 1. Introduction

The **Smart Traffic Prediction System** is a web-based intelligent platform designed to analyze, predict, and manage traffic flow efficiently in urban areas. It leverages advanced technologies such as Machine Learning and real-time data processing to provide accurate traffic forecasts and improve transportation planning. The system is developed using modern web frameworks and deployed on cloud infrastructure to ensure scalability, high availability, and secure access.

The platform allows users such as traffic authorities and commuters to access real-time traffic updates, congestion predictions, and route suggestions through a user-friendly interface. It integrates multiple data sources, including historical traffic data, to generate reliable predictions. Users can view traffic trends, receive alerts about potential congestion, and plan optimal routes accordingly.

Administrators can manage traffic data, monitor system performance, and configure prediction models, while the system ensures secure access through role-based authentication and authorization mechanisms. Data is securely stored in cloud databases, and large datasets are efficiently handled using cloud storage solutions.

Overall, the Smart Traffic Prediction System provides a centralized and intelligent solution that enhances traffic management, reduces congestion, improves travel efficiency, and supports smarter decision-making for modern cities.

## 2. Cloud Services Used

| Cloud Service                                   | Type | Purpose                                                                                                                                                  |
|-------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Streamlit (Cloud / Hosting)                     | PaaS | Provides the user interface for the Smart Traffic Prediction System, enabling interactive dashboards, real-time data visualization, and user interaction |
| Amazon Web Services EC2 (Elastic Compute Cloud) | IaaS | Hosts the entire application (Flask API, Streamlit UI, and ML model) and handles all processing                                                          |

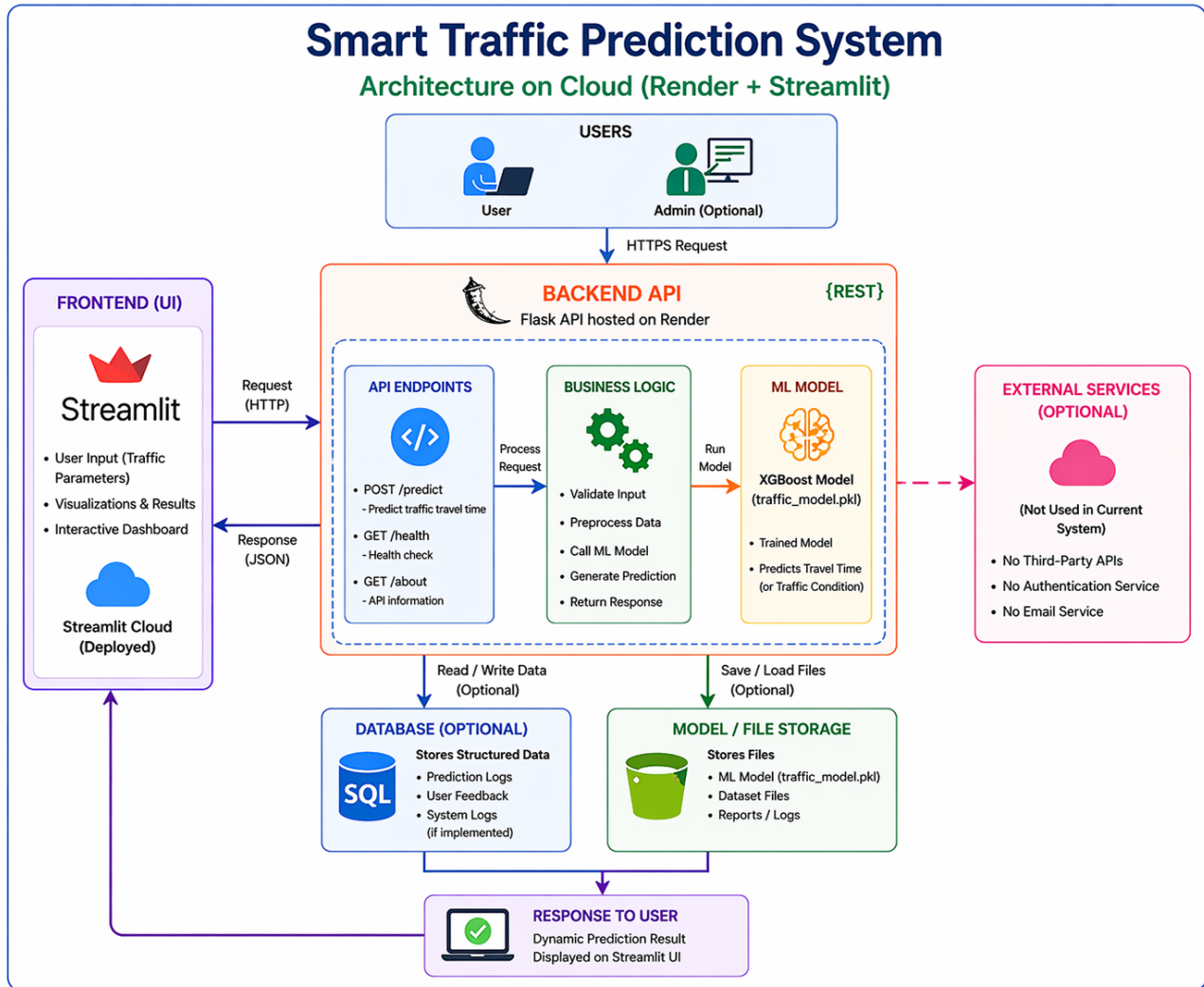
### External APIs (Cloud / Third-Party Services)

| External API    | Provider | Purpose                                                                   | Integrated With                     |
|-----------------|----------|---------------------------------------------------------------------------|-------------------------------------|
| Google Maps API | Google   | Provides real-time traffic data, route information, and map visualization | Backend API (Render) & Streamlit UI |

### **3. Cloud service Integration**

- 1. The system integrates AWS services in the following way:**
  - The application is deployed on an EC2 instance, which runs:
    - Flask backend API
    - Streamlit frontend
    - Machine learning model
- 2. Users access the system through a public EC2 URL via a web browser.**
- 3. The Streamlit UI interacts with the Flask API locally (within EC2) or via internal HTTP calls.**
- 4. The Flask API:**
  - a. Processes user input
  - b. Calls the ML model
  - c. Returns prediction results
- 5. This architecture ensures:**
  - Centralized deployment
  - Secure access
  - Scalable infrastructure

# 4. Architecture Design



# 5. Component Explanation

## 1. Frontend (Streamlit UI)

- Provides interactive web interface
- Accepts user inputs (traffic parameters)
- Sends requests to backend API
- Displays prediction results

## 2. Backend (Flask API)

- Handles incoming requests
- Validates and preprocesses input data
- Calls machine learning model
- Returns prediction as JSON response

## 3. Machine Learning Model (XGBoost)

- Trained on traffic dataset
- Stored as traffic\_model.pkl
- Predicts travel time or traffic condition

## 4. Cloud Infrastructure (AWS EC2)

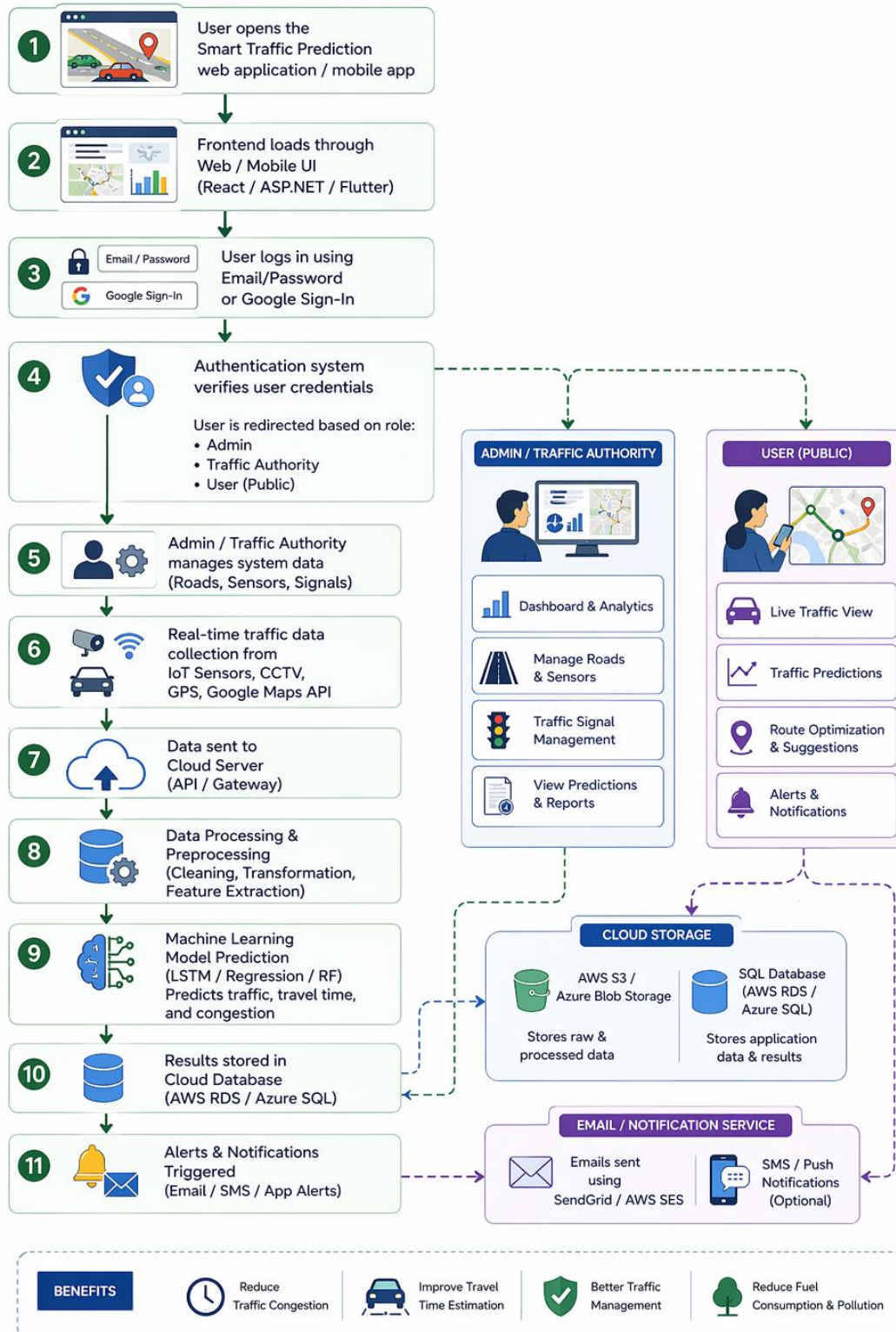
- Hosts complete system
- Runs application services
- Provides public access via IP/URL

## 5. Security (IAM)

- Controls access to AWS resources
- Ensures secure deployment

## 6. System Workflow

### System Workflow of Smart Traffic Prediction System



- User opens the Smart Traffic Prediction web application / mobile app
- Frontend loads through Web/Mobile UI (React / ASP.NET / Flutter interface)
- User logs in using Email/Password or Google Sign-In
- Authentication system verifies user credentials
- User is redirected based on role (Admin / Traffic Authority / Public User)
- Admin / Traffic Authority manages system data (roads, sensors, traffic signals)
- Real-time traffic data is collected from IoT sensors, CCTV cameras, GPS, and APIs (e.g., Google Maps)
- Collected data is sent to the cloud server via API / Gateway
- Data is processed and preprocessed (cleaning, transformation, feature extraction)
- Machine Learning model predicts traffic congestion, travel time, and traffic patterns
- Prediction results are stored in cloud database (AWS RDS / Azure SQL)
- Raw and processed traffic data is stored in cloud storage (AWS S3 / Azure Blob Storage)
- Alerts and notifications are sent to users (Email / SMS / App notifications)
- Public users view live traffic updates and predicted traffic conditions
- System provides optimized route suggestions based on predictions
- System continuously updates and improves predictions using new data

## 7. Conclusion

The Smart Traffic Prediction System successfully provides an efficient and intelligent platform for monitoring and managing traffic conditions in real time. It enables traffic authorities to collect and manage traffic data, analyze patterns, and make informed decisions, while allowing public users to view live traffic updates, predicted congestion levels, and optimized route suggestions.

The integration of technologies such as Machine Learning models for prediction, real-time data collection through IoT sensors and APIs, cloud storage (AWS S3 / Azure Blob Storage), and database management systems (AWS RDS / Azure SQL) ensures that the system is scalable, reliable, and secure. Additionally, notification services help keep users informed about traffic conditions and alerts.

Overall, the project demonstrates how cloud computing and data-driven approaches can significantly improve traffic management, reduce congestion, optimize travel time, and contribute to smarter and more sustainable urban transportation systems.

